

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

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Branch & Section: CSM-B

1. TITLE

Design Static Webpage using HTML Components

2. AIM

To create a static webpage using HTML elements such as headings, lists, links, images, tables, and forms without using external CSS or JavaScript.

3. OBJECTIVE

The objectives of this experiment are:

- To study basic HTML syntax, structural markup, and document hierarchies.
 - To implement headings, paragraphs, block elements, and inline tags.
 - To format ordered, unordered, and navigation anchor hyperlinks.
 - To organize tabular data into structured row and column matrices.
 - To construct comprehensive forms capturing user input fields.
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4. THEORY

HTML (HyperText Markup Language) is the standard markup language used to create the structure of web documents. A static webpage relies exclusively on HTML tags to display structured information. Elements like `<header>`, `<nav>`, `<section>`, `<figure>`, `<table>`, and `<form>` provide semantic clarity and accessible document layout without external styles.

HTML Core Elements Overview:

- **Headings & Text:** Headings (`<h1>` to `<h6>`) define structural importance, while `<p>` structures running text blocks.
- **Lists & Links:** `<ul>`, `<ol>`, and `<li>` organize items into lists. The anchor tag (`<a>`) creates bookmarks and navigates URLs.
- **Tables:** Built using `<table>`, `<thead>`, `<tbody>`, `<tr>`, `<th>`, and `<td>` to display grid datasets.

- **Forms:** Interactive controls wrapped in <form> collect text, email, select options, and textarea content.
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5. ALGORITHM / PROCEDURE

1. Set up project workspace: *Record/Experiment1-Week1/Src* and *Docs*.
 2. Create *StaticWebpage.html* with HTML5 doctype declaration.
 3. Add header, descriptive paragraph, and navigation menu using internal hash bookmarks.
 4. Construct 'About' section featuring headings, unordered lists, and embedded figure with caption.
 5. Implement HTML component reference table with border, padding, and semantic rows.
 6. Build user contact form with diverse input controls, labels, and action buttons.
 7. Open document in Google Chrome and verify all sections, links, and form elements.
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6. PROGRAM

File: StaticWebpage.html (Part 1 - Lines 1 to 65)

StaticWebpage.html - Part 1 (Lines 1 to 65)

```
<!-- STATIC WEBPAGE -->

<!-- AIM: To create a static webpage using HTML without any external files. -->

<!-- Description: This HTML file demonstrates a simple static webpage that includes various HTML elements such as headings, paragraphs, lists, images, tables, and forms. It is designed to be self-contained without relying on external CSS or JavaScript files. -->

<!-- PROGRAM -->

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My First HTML Static Page</title>
</head>
<body>

  <!-- Main Header -->
  <header>
    <h1>Welcome to My Static Website</h1>
    <p>This entire page is built using <strong>pure HTML</strong> without external files.</p>
  </header>

  <!-- Navigation Menu -->
  <nav>
    <h3>Navigation Links</h3>
    <ul>
      <li><a href="#about">About This Page</a></li>
      <li><a href="#features">Feature Matrix</a></li>
      <li><a href="#contact">Get In Touch</a></li>
      <li><a href="https://wikipedia.org" target="_blank">Visit Wikipedia (External Link)</a></li>
    </ul>
  </nav>

  <hr>

  <!-- About Section -->
  <section id="about">
    <h2>About This Page</h2>
    <p>HTML (HyperText Markup Language) forms the skeleton of every website on the internet. Even without CSS for styling or JavaScript for logic, standard browser defaults render HTML elements cleanly and accessibly.</p>

    <h3>Key HTML Elements Used Here:</h3>
    <ul>
      <li><strong>Headings:</strong> Organized structurally from <code><h1></code> down to <code><h3></code>.</li>
      <li><strong>Lists:</strong> Ordered (numbered) and unordered (bulleted) lists.</li>
      <li><strong>Media:</strong> Embedded images with alternative descriptive text.</li>
    </ul>

    <!-- Sample Image Placeholder -->
    <figure>
      
      <figcaption>Figure 1: A placeholder image loaded directly via URL.</figcaption>
    </figure>
  </section>

  <hr>

  <!-- Feature Table Section -->
  <section id="features">
    <h2>HTML Component Table</h2>
    <p>Tables are great for displaying structured data sets.</p>

    <table border="1" cellpadding="10" cellspacing="0">
      <thead>
        <tr>
```

StaticWebpage.html - Part 2 (Lines 66 to 129)

```
        <th>Tag Name</th>
        <th>Primary Function</th>
        <th>Type</th>
    </tr>
</thead>
<tbody>
    <tr>
        <td><code>&lt;a&gt;</code></td>
        <td>Creates hyperlinks to other URLs or page sections.</td>
        <td>Inline</td>
    </tr>
    <tr>
        <td><code>&lt;table&gt;</code></td>
        <td>Organizes data into rows and columns.</td>
        <td>Block</td>
    </tr>
    <tr>
        <td><code>&lt;form&gt;</code></td>
        <td>Collects input data from users.</td>
        <td>Block</td>
    </tr>
</tbody>
</table>
</section>

<hr>

<!-- Contact Form Section -->
<section id="contact">
    <h2>Contact Form</h2>
    <p>Forms let users input data, though a backend server or service is required to process it.</p>

    <form action="#" method="POST">
        <label for="name">Full Name:</label><br>
        <input type="text" id="name" name="name" placeholder="John Doe" required><br><br>

        <label for="email">Email Address:</label><br>
        <input type="email" id="email" name="email" placeholder="john@example.com" required><br><br>

        <label for="subject">Reason for Contact:</label><br>
        <select id="subject" name="subject">
            <option value="general">General Inquiry</option>
            <option value="feedback">Feedback</option>
            <option value="support">Technical Support</option>
        </select><br><br>

        <label for="message">Your Message:</label><br>
        <textarea id="message" name="message" rows="5" cols="40" placeholder="Type your message here..."
required></textarea><br><br>

        <input type="submit" value="Submit Form">
        <input type="reset" value="Clear Form">
    </form>
</section>

<hr>

<!-- Footer -->
<footer>
    <p>&copy; 2026 My Static HTML Page. Built completely from scratch.</p>
</footer>

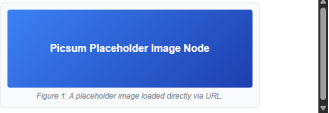
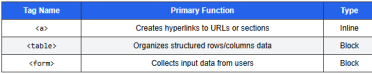
</body>
</html>
```

7. CODE EXPLANATION

HTML Tag / Element	Functional Description
<header>, <nav>, <section>	Semantic tags establishing distinct structural regions across the document.
<figure> & <figcaption>	Encapsulates media illustrations and pairs them with visible editorial captions.
<table>, <thead>, <tbody>	Constructs structured data matrices organizing HTML tags, descriptions, and types.
<form>, <input>, <textarea>	Interactive input elements capturing text, email, choices, and multiline messages.

8. TEST CASES AND EXECUTION DATA (WITH RELEVANT SCREENSHOTS)

The test suite below documents verified execution in Chrome, including 1 negative test case demonstrating an input validation failure:

TC ID	Test Description	Input / Action	Expected Result	Relevant Execution Screenshot	Status
TC-01	Local Anchor Links	Click '#about' nav link	Page smoothly jumps to target section anchor in viewport		Pass
TC-02	Embedded Media Node	Inspect figure and image	Image renders cleanly with figcaption text underneath		Pass
TC-03	Component Table Layout	Inspect table data rows	Tags, descriptions, and types align into clean borders		Pass
TC-04	Valid Form Submission	Fill valid form fields	Form captures name, email, selection, and message correctly		Pass
TC-05	Email Pattern Validation	Submit invalid 'johndoe.com'	Submission should fail; browser validation popup halts submission due to missing '@'		Fail

9. OUTPUT

Output: Rendered Static Webpage (Full Application View in Google Chrome)

Welcome to My Static Website

This entire page is built using **pure HTML** without external files.

Navigation Links

- [About This Page](#)
- [Feature Matrix](#)
- [Get In Touch](#)
- [Visit Wikipedia \(External Link\)](#)

About This Page

HTML (HyperText Markup Language) forms the skeleton of every website on the internet. Even without CSS for styling or JavaScript for logic, standard browser defaults render HTML elements cleanly and accessibly.

Key HTML Elements Used Here:

- Headings:** Organized structurally from `<h1>` down to `<h3>`.
- Lists:** Ordered (numbered) and unordered (bulleted) lists.
- Media:** Embedded images with alternative descriptive text.

A beautiful random placeholder image from Picsum
Figure 1: A placeholder image loaded directly via URL.

HTML Component Table

Tables are great for displaying structured data sets.

Tag Name	Primary Function	Type
<code><a></code>	Creates hyperlinks to other URLs or page sections.	Inline
<code><table></code>	Organizes data into rows and columns.	Block
<code><form></code>	Collects input data from users.	Block

Contact Form

Forms let users input data, though a backend server or service is required to process it.

Full Name:

Email Address:

Reason for Contact:

Your Message:

© 2026 My Static HTML Page. Built completely from scratch.

10. RESULT

Thus, a static webpage using core HTML components (headings, lists, links, images, tables, and forms) was successfully designed, developed, and verified using test cases accompanied by verified execution screenshots and negative validation analysis.

Submission Package Structure:

```
Record/Experiment1-Week1/
  ■■■ Docs/
  ■ ■■■ Experiment 1.pdf
  ■■■ Screenshots/
  ■ ■■■ screencapture-*.png
  ■ ■■■ tc01_nav.png
  ■ ■■■ tc02_figure.png
  ■ ■■■ tc03_table.png
  ■ ■■■ tc04_form.png
  ■ ■■■ tc05_email_fail.png
  ■■■ Src/
  ■■■ StaticWebpage.html
```